

RANSOMWARE

Critical	May, 23, 2021, 07:32 PM	SOC145 - Ransomware Detected	92	Malware
EventID :	92			
Event Time :	May, 23, 2021, 07:32 PM			
Rule :	SOC145 - Ransomware Detected			
Level :	Security Analyst			
Source Address :	172.16.17.88			
Source Hostname :	MarkPRD			
File Name :	ab.exe			
File Hash :	0b486fe0503524cfe4726a4022fa6a68			
File Size :	775.50 Kb			
Device Action :	Allowed			
File (Password:infected) :	Download			
Show Hint				

Ransomware – Type of cyberattack where actors take over a system by using malicious software to encrypt data/files in hopes of extorting the owners of the system i.e. financial reasons

- Crypto Ransomware = encrypts files on a victim's system
- Locker Ransomware = locks users out of their entire system

Associated resources/Devices

Time of attack = May, 23, 2021, 07:32 PM
Source Ip = 172.16.17.88
Source hostname = MarkPRD
File name = ab.exe
File hash = 0b486fe0503524cfe4726a4022fa6a68

1) Threat Indicator

Define Threat Indicator

Select Threat Indicator

Next

This defines a sign that indicates malicious activity:

Anti-virus programs malfunctioning or becoming disabled for unknown reasons

2) Quarantine

Check if the malware is quarantined/cleaned

- Log Management
- Endpoint Security

Malware quarantined/cleaned?

Not Quarantined Quarantined

The end-user device has not been quarantined (Not Quarantined) as seen on the endpoint security interface below.

Action

Containment: ☐

Further analysis to determine the extent of the attack can be done using:

Log Management

We can filter the logs based on the source Ip address

Show Filter		Search				
DATE	TYPE	SRC ADDRESS	SRC PORT	DEST. ADDRESS	DEST. PORT	RAW
Aug, 29, 2020, 11:09 PM	Proxy	172.16.17.88	23477	81.169.145.105	80	🔍
Apr, 04, 2021, 11:10 PM	Proxy	172.16.17.88	39483	192.64.119.190	80	🔍

RAW LOG	RAW LOG
Request URL: http://thuening.de/cgi-bin/uo9wm/ Request Method: GET Device Action: Permitted Process: powershell.exe Parent Process: BAL_GB9684140238GE.doc Parent MD5: ac596d282e2f9b1501d66fce5a451f00	Request URL: http://nuangaybantiep.xyz Request Method: GET Device Action: Allowed Process: chrome.exe Parent Process: explorer.exe Parent Process MD5: 8b88ebbb05a0e56b7dcc708498c02b3e

Lets determine if the source Ip(172.16.17.88) is malicious using resources like:

AbuseIPDB

AbuseIPDB » [172.16.17.88](#)

Check an IP Address, Domain Name, or Subnet
e.g. 41.90.187.91, microsoft.com, or 5.188.10.0/24

172.16.17.88

CHECK

172.16.17.88 was not found in our database

Important Note: 172.16.17.88 is a private IP address, and is only used in internal network environments. Any abusive activity you see coming from an internal IP is either coming from within your network itself, or is the result of an error or misconfiguration.

With this in mind, we present the reports on this page for entertainment and testing purposes only. If you mistakenly blacklist an internal IP, you will not have a good day!

IP Abuse Reports for 172.16.17.88

This IP address has not been reported. [File Report](#)

Using the above platform, we can determine that the Ip address is private i.e. Class B and has not been flagged by any engine.

Class A: 10.0.0.0 – 10.255.255.255 (Large organizations)

Class B: 172.16.0.0 – 172.31.255.255 (Medium-sized networks)

Class C: 192.168.0.0 – 192.168.255.255 (Home and small business networks)

Endpoint Security

MarkPRD	Endpoint Information
MarkPRD 172.16.17.88	Host Information Host Information Hostname: MarkPRD Domain: LetsDefend IP Address: 172.16.17.88 Bit Level: 64 OS: Windows 10 Primary User: MarkGuna Client/Server: Client Last Login: Aug, 29, 2020, 08:12 PM Action Containment: ☐

Using the source Ip (172.16.17.88) and hostname we can view the device compromised

Note:

Device Action :

Allowed

By this we can verify that the end user device was compromised thus action needs to be taken to prevent escalation of the attack within the network

Event Time : May, 23, 2021, 07:32 PM

There's a time difference between the time the attack was triggered and the last login of the user device

based on the endpoint info Last Login: Aug, 29, 2020, 08:12 PM

EVENT TIME	PROCESS ID	PROCESS NAME	PARENT PROCESS	COMMAND LINE
No Event Time	No Process ID	TiWorker.exe	—	C:/Windows/WinS...
No Event Time	No Process ID	ab.exe	—	No Command

MD5 : 0b486fe0503524cfe4726a4022fa6a68

We can verify the file (ab.exe) and hash function(0b486fe0503524cfe4726a4022fa6a68) to determine if they are malicious

VirusTotal

62/72

Community Score

62/72 security vendors flagged this file as malicious

taskhost.exe

Size: 775.50 KB

Last Analysis Date: 19 hours ago

DETECTION

Join our Community

Popular threat label: ransomware.avaddon/delshad

Threat categories: ransomware, trojan

Family labels: avaddon, delshad, smtha

Security vendors' analysis

AhnLab-V3	Malware/Win.Ransom.8421765	Alibaba	Ransom:Win32/Avaddon-4ec8be5
AliCloud	RansomWare/Win/Avaddon	ALYac	Trojan.Ransom.Avaddon
Antiy-AVL	Trojan(Ransom)/Win32.Avaddon	Arcabit	Trojan.Ransom.Avaddon.3
Arctic Wolf	Unsafe	Avast	Win32:MalwareX-gen (Ransom)
AVG	Win32:MalwareX-gen (Ransom)	Avira (no cloud)	HEUR/JAGEN.1311690
BitDefender	Gen:Variant.Ransom.Avaddon.3	Bitdefender	W32.Common.36033F49
ClamAV	Win.Ransomware.Avaddon-9852658-0	CrowdStrike Falcon	Win/malicious_confidence_100% (W)
CTX	Evx.ransomware.avaddon	Cymet	Malicious (score: 99)

Based on the above output we can ascertain that the file is corrupted as the hash has been flagged by multiple engines.

3) Malicious

Analyze Malware

Analyze malware in 3rd party tools and find C2 address

You can use the free products/services below.

- AnyRun
- VirusTotal
- URLHouse
- URLScan
- HybridAnalysis

Malicious Non-malicious

By the tests done earlier we can determine the file is **malicious** as multiple engines flagged the hash

There is a need to test using multiple services to ensure optimum results

Command-and-Control (C2) server = remote system used by attackers to communicate with malware-infected devices

4) Request C2

Check If Someone Requested the C2

Please go to the "Log Management" page and check if the C2 address accessed. You can check if the malicious file is run by searching the C2 addresses of the malicious file.

- Log Management

Please click "Accessed" if someone access the malicious address. Otherwise please click "Not Accessed" button.

The malicious address has not been accessed (**NOT ACCESSED**) based on the logs below as neither the file nor the hash is recorded

Show Filter		Search				
DATE	TYPE	SRC ADDRESS	SRC PORT	DEST. ADDRESS	DEST. PORT	RAW
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Apr, 04, 2021, 11:10 PM	Proxy	172.16.17.88	39483	192.64.119.190	80	🔍

Also as shown below no log data/ commands are recorded

⬆ No Event Time	No Process ID	ab.exe	—	No Command
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5) Artifacts

Add Artifacts

+

Value	Comment	Type	Remove
172.16.17.88	end user device targeted	IP Address	🗑
0b486fe0503524cfe4726a4022fa6a68	hash of the file	MD5 Hash	🗑

Artifacts = Indicators of Compromise

IP Address: 172.16.17.88 -The targeted end-user device within the private network.

MD5 Hash: 0b486fe0503524cfe4726a4022fa6a68 -The malicious file's unique identifier.

6) Analysis Note

Analyst Note

Please enter your analysis comments.

The attack is a true positive as the hash and file were flagged by multiple services as malicious. The first step to counter the attack is to contain the device to prevent/ stop further escalation of the attack within the network

230 / 3000

Key Threat Indicators:

Antivirus malfunctioning, possibly disabled by malware.

Unexpected network traffic (potential C2 communication).

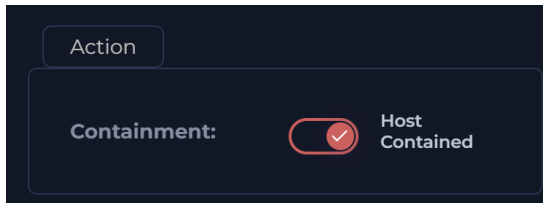
Possible system modifications or persistence mechanisms.

7) Close Case

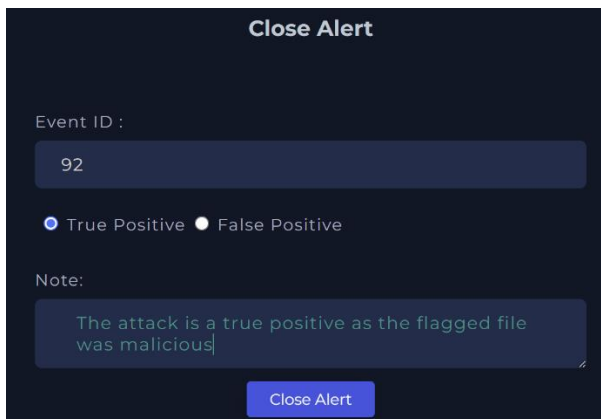
After analysing the attack, we need to mitigate:

First, contain the device to prevent lateral movement and further damage.

-Quarantine affected device



Close the alert

A dark-themed 'Close Alert' form. The title 'Close Alert' is at the top. Below it, the label 'Event ID :' is followed by a text input field containing the number '92'. Underneath, there are two radio buttons: 'True Positive' (which is selected) and 'False Positive'. A 'Note:' label is followed by a text area containing the text 'The attack is a true positive as the flagged file was malicious'. At the bottom right, there is a blue button labeled 'Close Alert'.